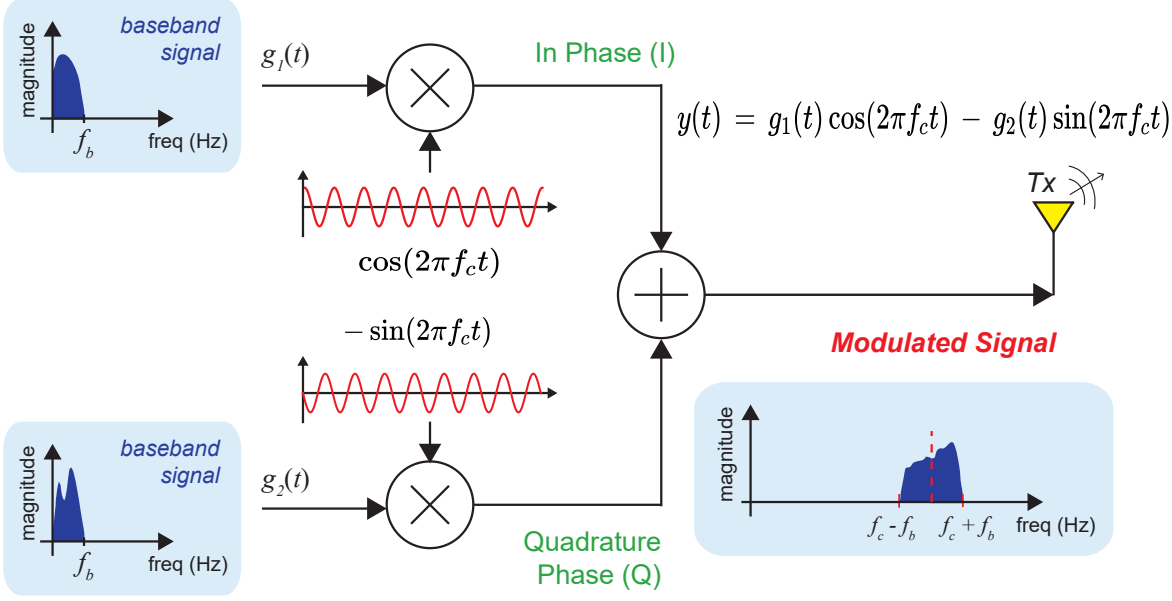

baseband signal
magnitude
f_b
freq (Hz)
$g_1(t)$
In Phase (I)
$y(t) = g_1(t)\cos(2\pi f_c t) - g_2(t)\sin(2\pi f_c t)$
Tx
$\cos(2\pi f_c t)$
$-\sin(2\pi f_c t)$
Modulated Signal
baseband signal
magnitude
f_b
freq (Hz)
$g_2(t)$
Quadrature Phase (Q)
magnitude
$f_c - f_b$
$f_c + f_b$
freq (Hz)